sPHENIX Self-Paced Course — Module 10

Case studies from the 2026 school and machine learning in sPHENIX

Prepared for P. D. Lewis (Ohio University) November 2026

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How to use this module

This module is self-contained. It does not assume that you have seen the slides or recordings of the school. Each case study explains the physics from first principles, then the method, then what it teaches you. The talks are listed as *sources*, cited [S##]; all numbers quoted are public design values, textbook or PDG values, or explicitly labelled toy numbers. Results shown at the school that are internal or preliminary are described only qualitatively.

Learning objectives. After this module you will be able to:

- take any sPHENIX analysis apart into question, observable, data, method, key challenge and lessons;
- explain the physics and method of sixteen student analyses spanning all four topical groups and detector work;
- recognise the recurring tools (mixing and rotation backgrounds, ABCD purity, tag-and-probe, reference spectra, closure tests);
- set up a supervised ML classifier or regressor for a physics task, choose its loss, validate it on data and assign systematics;
- write a two-page capstone proposal tied to your π^0/η cross-section analysis.

Prerequisites. All earlier modules; Module 9 is used most. Basic Python is helpful for the ML part.

Estimated time. 14 hours (two weeks), plus the capstone proposal.

Sources (sPHENIX School 2026 flash talks). [S26] C.-W. Shih; [S25] H.-R. Jheng; [S34] I. Omae; [S37] J. Singh; [S39] L. Mwibanda; [S38] K. Gable; [S35] R. Takaaze; [S40] M. I. Abdulhamid; [S29] B. Sayki; [S32] S.-Y. Chen; [S30] J. Hwang; [S31] R. Koike; [S41] T. Harada; [S27] A. Patton; [S28] T. Kikuchi; [S42] A. M. Khan. Supporting talks: [S05] Bulk TG, [S04] Cold QCD TG, [S17] Photon reconstruction, [S43] Jet analysis. Full titles are in the references.

Public readings. sPHENIX papers on Au+Au multiplicity (arXiv:2504.02240), transverse energy (arXiv:2504.02242), dijet imbalance (arXiv:2606.17184) and isolated prompt photons (arXiv:2607.03892).

1 How to read an analysis

Goal. Learn a fixed template that turns any talk, note or paper into six answers. You will use it for every case study and for your own capstone.

1. **Question.** What physics question is asked? Which theory quantity does the answer constrain?
2. **Observable.** What is measured, defined precisely enough to compute in a generator?
3. **Data.** Collision system, run period, trigger, readout mode (triggered or streaming), detectors required, simulation.
4. **Method.** Reconstruction, signal extraction, background estimation, corrections, normalisation.
5. **Key challenge.** The one step that limits precision or validity.
6. **Lesson for you.** What transfers to your π^0/η cross section or R_{CP} work.

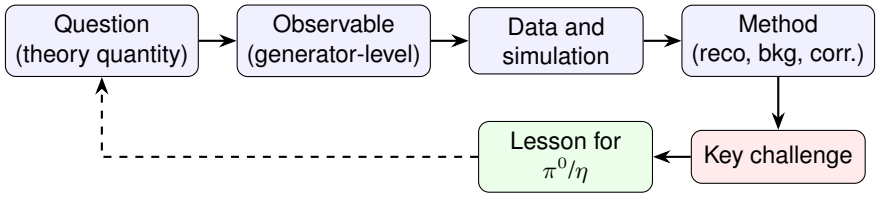


Figure 1 (schematic): the six-question template. The dashed arrow: lessons change the questions you ask next.

Three recurring ideas appear in nearly every case below. Look for them.

- **Destroy the signal, keep the detector.** Event mixing, layer rotation and like-sign pairs all estimate combinatorial background by breaking physical correlations while keeping acceptance.
- **Use a reference.** A summed ring spectrum, an average run, a second method, or an independent detector provides the baseline against which a deviation is measured.
- **Close the loop in simulation.** Every correction is applied to simulated events treated as data, and must give back the truth.

2 Bulk physics case studies

2.1 Charged-hadron multiplicity in O+O with INTT tracklets [S26]

Physics. The pseudorapidity density of charged particles, $dN_{\rm ch}/d\eta$, measures how much entropy the collision produces. In the Bjorken picture it also sets a lower bound on the initial energy density (Module 1). Its scaling with $N_{\rm part}$ across system size tests particle-production models. O+O is the first light symmetric ion system at RHIC, and the LHC has measured O+O at TeV energies, so a RHIC measurement fills a gap in energy [S25, S26].

Method. A *tracklet* is a pair of clusters, one in each INTT barrel layer, consistent with coming from the event vertex. For each pair the azimuthal difference $\Delta\phi$ is histogrammed. With the magnetic field on, positive and negative particles bend opposite ways, so the true-pair peak in $\Delta\phi$ has two components. Random pairs are estimated by rotating the inner layer by π and repeating the pairing; the rotated distribution is subtracted and the signal counted inside a fixed $|\Delta\phi|$ window. A simulation-derived factor $f(\eta, \mathrm{cent}) = N_{\rm reco}/N_{\rm truth}$ converts tracklets to primary charged hadrons. The MC vertex distribution is reweighted to data. Without an official centrality calibration, centrality was obtained from an NBD-convolved Glauber fit to the MBD charge sum (Module 9). The INTT operated in triggered mode; a hit-carry-over effect was removed by an event-interval cut, and a bunch-crossing rule (use only the later crossing of an adjacent pair) avoided collisions recorded through the extended readout window. That rule depends only on crossing number, so it adds no centrality bias. Systematics came from varying cluster ADC and ϕ-size cuts, the $\Delta\phi$ window, and the offline geometry (survey only versus with alignment).

Question	How does particle production per participant pair scale in a small symmetric system at RHIC energy?
Observable	$dN_{\rm ch}/d\eta$ for $\lvert\eta\rvert < 1.1$ versus centrality, normalised by $\langle N_{\rm part}\rangle/2$.
Data	Run-25 O+O, INTT in triggered mode, MBD for centrality and vertex; HIJING through Geant4.
Method	INTT doublets, rotation background, MC correction factor, NBD–Glauber centrality.
Key challenge	Centrality and $N_{\rm part}$ in a small system, and generator artifacts (an unexpected structure in one official HIJING $N_{\rm part}$ distribution).
Lesson	Validate the generator at the level you use it; make the MC match the data in vertex and dead channels; vary detector-level cuts for systematics.

2.2 The same measurement with silicon seeds [S25]

Physics and method. The standard sPHENIX tracking first builds *silicon seeds* from MVTX and INTT clusters (Module 8). Seeds are labelled by cluster counts, e.g. "3+2" (three MVTX, two INTT), "3+1", "2+2", "2+1". Seeding deliberately over-produces candidates to maximise efficiency; fakes are normally removed when seeds are matched to the TPC and fitted. Used directly for counting, raw seeds therefore overcount. [S25] compared the seed multiplicity with INTT doublets and with truth, in simulation with pile-up and in data, and studied two handles: a vertex-association cut on the seed's point of closest approach to the beam line, and a pruning algorithm that removes seeds sharing clusters. Shapes in η agreed between data and simulation for each seed class, but the overall normalisations differed, which is exactly the information needed for a correction. A survey of heavy-ion generators showed widely different predictions for O+O at 200 GeV, so the measurement will discriminate between models.

Question	Can a standard reconstruction object (silicon seed) deliver $dN_{\rm ch}/d\eta$, cross-checked by an independent method?
Observable	$dN_{\rm ch}/d\eta$ from seed counts; seed-to-doublet and seed-to-truth ratios.
Data	O+O data and HIJING with realistic pile-up; data dead-channel map applied to simulation.
Method	Seed classification, vertex association, pruning; comparison with INTT doublets.
Key challenge	Fake and duplicate seeds; selection that removes fakes also removes some good seeds.
Lesson	Two methods with different systematics that agree are the strongest evidence of control (the published Au+Au result used two methods [S05]). For you: an $\eta \to \gamma\gamma$ and $\pi^0 \to \gamma\gamma$ energy-scale check is such a pair.

2.3 Elliptic flow v_2 in Au+Au with INTT and MBD [S34]

Physics. In a non-central collision the overlap region is almond-shaped. Hydrodynamic pressure gradients are steeper along the short axis, converting spatial eccentricity ε_2 into momentum anisotropy $v_2 = \langle \cos 2(\phi - \Psi_2) \rangle$. The ratio v_2/ε_2 is sensitive to the shear viscosity over entropy density of the QGP [S05].

Method. The event plane Ψ_2 was measured with one detector and particles with another, separated in η: INTT particles with an MBD plane, and MBD particles with an INTT plane. The plane was corrected by recentering and Fourier flattening; the resolution was obtained from three sub-events (INTT, MBD south, MBD north) and divided out. Systematics: change the plane detector and the INTT ADC threshold. The η dependence was compared with earlier PHOBOS data (Module 9, Section 5.3, gives the formulas).

Question	How large is collective flow and how does it depend on centrality and η?
Observable	v_2 versus centrality and η.
Data	Run-24 Au+Au, field off, MB trigger, $\lvert z \rvert$ cut.
Method	Event plane with recentering, flattening, three-sub-event resolution.
Key challenge	Detector non-uniformity and non-flow; low-p_T and background effects, to be studied with simulation.
Lesson	Each correction has a diagnostic plot (Ψ flatness, χ^2/ndf of a constant fit, resolution versus centrality). Your π^0 v_2 would use the same plane.

2.4 Constraining the chiral magnetic effect with the sliding dumbbell method [S37]

Physics. In non-central collisions the spectator protons create a very strong, short-lived magnetic field $\vec{B}$, perpendicular to the reaction plane. If topological gluon fields create an imbalance between right- and left-handed quarks (axial chemical potential μ_5), an electric current flows along $\vec{B}$: $\vec{J} = \frac{e^2}{2\pi^2}\mu_5\vec{B}$. This is

the chiral magnetic effect (CME). It would separate positive and negative charges across the reaction plane. The standard observable is the three-particle correlator

$$\gamma_{ab} = \langle \cos(\phi_a + \phi_b - 2\Psi_{RP}) \rangle, \qquad \Delta\gamma = \gamma_{OS} - \gamma_{SS}, \tag{1}$$

where OS and SS denote opposite- and same-sign pairs. A CME signal gives $\Delta\gamma > 0$, but so do backgrounds: resonance decays and local charge conservation coupled to elliptic flow give a term proportional to v_2/N. Isobar collisions (Ru+Ru versus Zr+Zr) were designed to have the same background and different $\vec{B}$; they did not show the expected difference.

Method. The sliding dumbbell method (SDM) looks event by event for charge separation. Two back-to-back azimuthal sectors (sides a and b, each 90° wide) are rotated in 1° steps. For each position compute $D_{b\pm} = \frac{n_a^+}{n_a^+ + n_a^-} + \frac{n_b^-}{n_b^+ + n_b^-}$ and keep the maximum, subject to a balance condition on the positive excess in a and negative excess in b. The fractional charge separation is $f_{DbCS} = D_{b\pm}^{\max} - 1$. Events are binned in f_{DbCS} percentiles; the top bins are enriched in CME-like events, and $\Delta\gamma$ is computed in each bin. [S37] studied this with the anomalous-viscous-fluid-dynamics (AVFD) model and with charge-shuffled samples as a background reference. The speaker also worked on TPC central-membrane laser calibration (Module 8), checking lamination fits run by run.

Question	Is there a CME contribution to charge separation, and can event selection enhance it?
Observable	$\Delta\gamma$ in classes of event-wise charge separation f_{DbCS}.
Data	Model events now; sPHENIX Au+Au with TPC tracks later.
Method	Sliding dumbbell scan; comparison with charge-shuffled and background-only samples.
Key challenge	Backgrounds that mimic the signal (local charge conservation); the result depends on modelling them.
Lesson	When signal and background give the same sign, a background-only reference sample (shuffling, mixing) is the measurement.

2.5 Ultra-peripheral ρ^0 photoproduction [S39]

Physics. When two nuclei pass with impact parameter larger than twice the nuclear radius, they do not interact hadronically. The strong electromagnetic field of each nucleus is a flux of quasi-real photons with intensity $\propto Z^2$. A photon can fluctuate into a $q\bar{q}$ pair and scatter elastically off the other nucleus, emerging as a vector meson (ρ^0, J/ψ). In *coherent* production the nucleus stays intact and the meson p_T is limited by the nuclear form factor to $p_T \lesssim \hbar c/R_A$, of order 0.1 GeV for Au. *Incoherent* production (nucleus breaks up) gives broader p_T. The $\rho^0 \to \pi^+\pi^-$ decay (mass about 775 MeV) gives a clean two-track event with almost nothing else.

Method. [S39] made projections with the STARlight generator for coherent and incoherent ρ^0 (and J/ψ), propagated through the full sPHENIX Geant4 simulation, with HIJING events as hadronic background. Like-sign pairs estimate the combinatorial background. UPC events are best collected in streaming readout, since they produce too little energy for standard triggers. In Run-25 Au+Au only some tracking detectors were streamed, so a dedicated INTT+TPC tracking would be needed.

Question	Can sPHENIX measure coherent and incoherent vector-meson photoproduction (nuclear gluon structure)?
Observable	ρ^0 and J/ψ yields versus p_T and rapidity.
Data	Streaming-readout O+O and Au+Au; STARlight and HIJING simulation.
Method	Two-track invariant mass; like-sign background; coherent/incoherent separation by p_T.
Key challenge	Tracking with partial detector sets in streaming mode; non-collision backgrounds.
Lesson	Feasibility studies with full simulation *before* data define what reconstruction must be developed.

2.6 ϕ-meson–hadron angular correlations in Au+Au [S38]

Physics. Strangeness enhancement (more strange hadrons per pion than in p+p) was an early QGP signature. ALICE found that the enhancement in small systems is largest in the underlying event and smaller in jets. The $\phi(1020)$, an $s\bar{s}$ state (mass 1019.5 MeV, width 4.2 MeV; PDG), has hidden strangeness and decays to K^+K^- about half the time.

Method. Choose a high-p_T trigger hadron; pair it with ϕ candidates reconstructed from K^+K^- pairs; histogram $\Delta\varphi = \varphi_{\rm trig} - \varphi_{\rm assoc}$. The near side ($\Delta\varphi \approx 0$) is the trigger jet, the away side ($\Delta\varphi \approx \pi$) the recoil jet that crossed the medium, and the flat pedestal the underlying event. Correct pair acceptance with mixed events, and compare per-trigger ϕ yields with those of inclusive charged hadrons in each region and multiplicity bin. The first step shown was PYTHIA $\phi \to K^+K^-$ reconstruction; the student also took on TPC central-membrane lamination-fit QA.

Question	Where in the event (jet or medium) is strangeness enhanced at RHIC?
Observable	Per-trigger ϕ/h yield ratios on near side, away side and underlying event, versus multiplicity.
Data	Au+Au with TPC tracking and kaon identification; PYTHIA and later Au+Au simulation.
Method	K^+K^- invariant mass; $\Delta\varphi$ correlation; mixed-event acceptance correction; flow-modulated background.
Key challenge	Large combinatorial kaon background in Au+Au; flow-modulated underlying event.
Lesson	Correlation functions use the same mixed-event logic as your π^0 background; per-trigger normalisation removes luminosity.

3 Jet case studies

3.1 First steps in a jet analysis [S35]

Physics. A jet is the collimated spray of hadrons from a hard-scattered parton. Jet algorithms must be infrared and collinear safe so that the result can be computed in perturbative QCD. The anti-k_T algorithm uses $d_{ij} = \min(p_{T,i}^{-2}, p_{T,j}^{-2})\,\Delta R_{ij}^2/R^2$ and $d_{iB} = p_{T,i}^{-2}$: it clusters soft particles around hard ones, giving cone-like jets of radius R (Module 7).

Method. [S35] described an onboarding sequence: set up the SDCC environment; read a p+p DST containing reconstructed jets; access the jet node; plot p_T, η, ϕ; and understand the chain (towers, retowering, anti-k_T, calibration). The student's background was in silicon-sensor test beams (ALICE FoCal), so the goal was to learn the reconstruction chain before a physics analysis combining calorimeter and tracking.

Question	How does one move from detector work to a physics analysis?
Observable	Basic jet kinematics, as a framework exercise.
Data	Run-24 p+p jet DSTs.
Method	Fun4All analysis module reading the jet container; histograms.
Key challenge	Learning the reconstruction chain and its calibrations, not the plotting.
Lesson	The same onboarding list works for any object; for you, the analogous chain is towers, clusters, pairs and mixing. Writing it down for a new student is a good test of your own understanding.

3.2 Joint jet mass and jet charge for quark/gluon separation [S40]

Physics. Quark and gluon jets differ because gluons carry colour charge $C_A = 3$ versus $C_F = 4/3$ for quarks: gluons radiate about $C_A/C_F = 9/4$ times more, so gluon jets are broader, have more constituents

and larger mass. The jet mass satisfies

$$m^2 = \Big(\sum_i E_i\Big)^2 - \Big|\sum_i \vec{p}_i\Big|^2 \approx p_T^2 \sum_{i<j} z_i z_j \, \Delta R_{ij}^2, \qquad z_i = p_{T,i}/p_{T,\mathrm{jet}}, \tag{2}$$

in the small-angle limit, so m/p_T measures jet width. The jet charge $Q_k = \sum_{i\in\mathrm{jet}} q_i \, (p_{T,i}/p_{T,\mathrm{jet}})^k$ is statistically sensitive to the charge of the initiating quark (up versus down); for gluons it averages to zero. The free parameter k (typically 0.3–1) weights hard constituents more for larger k.

Method. Calorimeter jet mass was studied in PYTHIA8 with full simulation, with unfolding closure tests (full and half). Tracks were associated to jets within $\Delta R < 0.3$ with quality cuts, giving jet charge. Since tracks have better angular resolution but miss neutrals, a *track-assisted mass* $m_{TA} = m_{\mathrm{track}}\, p_T^{\mathrm{calo}}/p_T^{\mathrm{track}}$ was planned. Photon-tagged jets are quark-enriched because Compton scattering $qg \to q\gamma$ dominates direct-photon production at RHIC, which links this study to the public isolated-photon measurement.

Question	Do jet mass and charge together separate quark from gluon jets at RHIC energies better than either alone?
Observable	Joint distribution of m (or m/p_T) and Q_k in inclusive and photon-tagged jets.
Data	PYTHIA8 with Geant4; later Run-24 p+p.
Method	Calorimeter mass, track association, jet charge, track-assisted mass, unfolding.
Key challenge	Calorimeter-only mass resolution; tracking efficiency inside jets.
Lesson	Combining tracking and calorimetry beats either alone. For your photons: a track veto or track isolation can remove charged contamination that the EMCal cannot.

4 Heavy-flavour case studies

4.1 J/ψ quarkonium energy correlators in p+p [S29]

Physics. Quarkonium production is described by NRQCD factorisation,

$$\sigma(pp \to H + X) = \sum_n \hat{\sigma}(pp \to Q\bar{Q}[n] + X)\, \langle \mathcal{O}_n^H \rangle, \tag{3}$$

where $\hat{\sigma}$ is perturbative (the heavy-quark mass $m_Q \gg \Lambda_{\mathrm{QCD}}$ sets a hard scale) and the long-distance matrix elements $\langle \mathcal{O}_n^H \rangle$ (LDMEs) describe the non-perturbative transition of a $Q\bar{Q}$ pair in state $n = {}^{2S+1}L_J^{[c]}$ (colour singlet or octet) to the hadron. Fits of LDMEs to cross sections disagree because different n give similar p_T spectra. A quarkonium energy correlator,

$$\Sigma(\cos\chi) = \int d\sigma \sum_i \frac{E_i}{M}\, \delta(\cos\chi - \cos\theta_i), \tag{4}$$

measures the energy flow around the quarkonium direction. Colour-octet states radiate before binding, so the radiation pattern differs by production channel, which can break the degeneracy.

Method and feasibility. sPHENIX has no muon system, so $J/\psi \to e^+e^-$ (branching ratio about 6%) is used. The expected yield factorises exactly like a cross section:

$$N_{J/\psi}^{\mathrm{obs}} = N_{\mathrm{MB}}\, \frac{\sigma_{J/\psi}}{\sigma_{\mathrm{MBD}}}\, \varepsilon_{\mathrm{MBD}}^{J/\psi}\, B_{ee}\, A\, \varepsilon_{\mathrm{trig}}\, \varepsilon_{\mathrm{reco}}. \tag{5}$$

Electrons are identified by $E_{\mathrm{EMCal}}/p \approx 1$ and little HCal energy. Pure training and validation samples come from data: photon conversions $\gamma \to e^+e^-$ (small opening angle) give electrons, and $K_S^0 \to \pi^+\pi^-$ (displaced vertex) gives pions. The 3 GeV photon trigger, with its turn-on, selects the events.

Question	Which NRQCD production channels dominate J/ψ production at RHIC?
Observable	Energy-weighted angular distribution of particles around the J/ψ.
Data	Run-24/25 p+p with tracking, EMCal and HCal; PYTHIA with full simulation; photon trigger.
Method	Electron ID (E/p, HCal veto), dielectron mass, energy correlator.
Key challenge	Pion contamination for electron p_T of 1–2 GeV; needs TPC calibration, alignment, track–calo matching all working.
Lesson	The feasibility formula is your cross-section formula read backwards; your $\varepsilon_{\mathrm{MBD}}$ for signal events and photon trigger turn-on are needed here too.

4.2 Muon identification with a CNN, INTT chip studies, and K^0_S spin asymmetry [S32]

Physics. A muon is a minimum-ionising particle: in the calorimeters it leaves a narrow track of small, nearly constant energy (roughly 2 MeV per g/cm^2). A charged pion of the same momentum usually starts a hadronic shower in the HCal. With no muon system, a calorimeter-based muon ID would open $J/\psi, \Upsilon \to \mu^+\mu^-$. Separately, the K^0_S is the CP-even combination $(K^0 - \bar{K}^0)/\sqrt{2}$, decays to two pions, and has $c\tau \approx 2.7$ cm (PDG), giving a clean displaced-vertex signal for a spin asymmetry sensitive to strange quarks.

Method. (i) A convolutional network took small calorimeter images around the track projection (EMCal and both HCal layers) plus scalar track variables (p_T, η, DCA, cluster counts) and output a muon score; trained on single-particle guns; performance was strongly p_T-dependent and was then tested on PYTHIA dimuon samples. (ii) INTT readout studies: the FPHX chip sets a "FIFO full" flag when its output buffer overflows; one chip with hot channels produced most flags, and a bad-channel mask from a Gaussian fit to channel hit rates removed them. A saturation of hits per chip was traced to a chip-level effect. (iii) First K^0_S reconstruction with KFParticle in streaming p+p data, and ϕ distributions toward A_N.

Question	Can muons be identified without a muon system? Is the INTT readout healthy? What is A_N for K^0_S?
Observable	Muon/pion rejection; flag rates; K^0_S azimuthal yields by spin state.
Data	Particle guns, PYTHIA; Run-24 Au+Au INTT data; streaming p+p for K^0_S.
Method	CNN plus dense network; hit-rate masks; KFParticle vertexing.
Key challenge	Muon ID performance at low p_T; validating ML on data.
Lesson	Hot-channel masking from a rate distribution is the INTT version of EMCal hot-tower masks in your run QA.

5 Cold QCD and photon case studies

5.1 Transverse single-spin asymmetry of isolated photons [S30]

Physics. In collisions of a transversely polarised proton with an unpolarised one, $A_N = (\sigma_L - \sigma_R)/(\sigma_L + \sigma_R)$. Large A_N for forward hadrons is attributed to initial-state effects (spin–k_T correlations, Sivers-type, or quark–gluon and three-gluon correlators in the collinear twist-3 picture) and final-state effects in fragmentation (Collins-type) [S04]. Direct photons have no fragmentation, so they isolate initial-state effects; at mid-rapidity Compton $qg \to q\gamma$ dominates, so they probe gluon correlations, in particular the tri-gluon correlator.

Method. The raw asymmetry is extracted with relative luminosity or the square-root formula and the $\sin\phi$ modulation, divided by the beam polarisation, and corrected for background (Module 9, Section 4.5):

$$A_N = \frac{A_N^{\mathrm{raw}} - r\, A_N^{\mathrm{bkg}}}{1 - r}, \qquad 1 - r = \text{purity}. \tag{6}$$

Purity comes from the ABCD method with signal-leakage correction (Module 9, Section 8), using a shower-shape BDT and isolation. The main background asymmetry, from π^0 and η decay photons, is known to be small. Extending to low p_T required re-training the BDT, re-validating ABCD independence, and a π^0/η *tag veto*: reject a photon that forms a pair in the π^0 or η mass window with any other cluster above a minimum energy.

Worked example: why the tag veto misses asymmetric decays

For a π^0 of energy E, the softer photon has $E_2 = \frac{E}{2}(1-\alpha)$ with α uniform on $[0,1]$. The partner is below a threshold $E_{\rm th}$ if $\alpha > 1 - 2E_{\rm th}/E$, which has probability $2E_{\rm th}/E$. For $E = 6$ GeV and $E_{\rm th} = 0.5$ GeV this is $1/6 \approx 17\%$, and those are exactly the decays in which the hard photon carries almost all the energy and looks like a single photon. This is why the student found that the remaining background is dominated by high-asymmetry π^0 with a soft partner, and that lowering the partner threshold is the lever.

Question	What is the gluon contribution to transverse-spin effects (tri-gluon correlator)?
Observable	A_N of isolated prompt photons at mid-rapidity versus p_T.
Data	Transversely polarised Run-24 p+p, photon triggers; PYTHIA.
Method	Square-root or relative-luminosity asymmetry; BDT photon ID; isolation; ABCD purity; tag veto.
Key challenge	Purity at low p_T, where π^0 background is largest and ABCD assumptions may fail.
Lesson	Your π^0 reconstruction *is* this analysis's background model. Your pair efficiency versus asymmetry and partner energy directly sets the tag-veto efficiency.

5.2 Neural-network discrimination of isolated photons and merged π^0 [S31]

Physics. At high energy the two π^0 photons merge into one cluster (Module 9: $\theta_{\rm min} \approx 2m_\pi/E$). The merged cluster is slightly elongated, with two sub-maxima, whereas a single photon gives a round shower. At low p_T, decay photons outnumber direct photons by more than an order of magnitude, so even a high-efficiency veto leaves signal-to-background of order one.

Method. The current photon ID uses a BDT on hand-crafted shower moments (for example the energy-weighted width $w_\eta = \sum_i E_i(\eta_i - \bar{\eta})^2 / \sum_i E_i$ and the core-energy fraction) [S17]. [S31] tried a network on raw tower information: each tower is a small feature vector ($\Delta\eta$, $\Delta\phi$ from the cluster centre, E, $E/E_{\rm cluster}$); a shared small network embeds each tower; a masked sum over towers gives a fixed-length, order-independent cluster representation; this is concatenated with cluster-level variables (η, E, number of towers) and passed to a classifier with two outputs. Training used cross-entropy on balanced single-photon and single-π^0 gun samples with one cluster per event, split into training, validation and test sets. On gun samples the separation was very good (high ROC AUC); the student stressed that it must still be validated on PYTHIA and on data control samples. See Section 8 for the general method.

Question	Can a network on raw towers beat a shower-shape BDT for γ/merged-π^0 separation?
Observable	ROC curve, signal efficiency at fixed background rejection.
Data	Single-particle gun simulation; later PYTHIA and data.
Method	Per-tower embedding, masked sum pooling, MLP classifier, cross-entropy loss.
Key challenge	Simulation-to-data transfer: real shower shapes, noise and calibration differ from simulation.
Lesson	Directly relevant to your high-p_T π^0: the same network can be trained to *select* merged π^0 clusters.

6 Detector, calibration and luminosity case studies

6.1 Vernier scans in Run-25 p+p and O+O [S41]

Physics and method. Module 9, Section 4 derived the scan: the rate versus beam separation is a Gaussian of width $\Sigma = \sqrt{\sigma_1^2 + \sigma_2^2}$, so with beam intensities one obtains $\mathcal{L}_0$ and $\sigma_{\mathrm{MBD}} = R_{\mathrm{MBD}}/\mathcal{L}_0$. [S41] analysed MBD and ZDC scans, applied beam-intensity and pile-up/accidental corrections (the latter by inclusion–exclusion of Poisson probabilities), used a double Gaussian for the ZDC because of tails, compared correction factors with the Run-24 analysis as a code and detector-stability check, and listed the hourglass correction as remaining work. The O+O cross section was smaller than in Au+Au, as expected from the smaller nuclei.

Question	What are the MBD and ZDC cross sections for each run period and system?
Observable	Coincidence rate versus beam separation.
Data	Dedicated scans in Run-25 p+p and O+O; accelerator beam-intensity data; GL1 scalers.
Method	Gaussian (single or double) fits per direction; intensity, pile-up, accidental and hourglass corrections.
Key challenge	Corrections for non-Gaussian shapes and the hourglass effect; comparison of two detectors.
Lesson	The σ_{MBD} in your luminosity comes from exactly this analysis; its uncertainty is your global normalisation uncertainty.

6.2 INTT hit-detection efficiency with streaming data [S28]

Physics and method. The efficiency of a detector layer can be measured in data by *tag-and-probe*: reconstruct tracks without the layer under test, project them onto it, and ask how often a hit is found nearby. [S28] used tracks built from the three MVTX layers plus the outer INTT layer to probe the inner INTT layer. Purity cuts mattered: the projected point had to be well inside a chip and away from dead or hot channels; the track had to come from the primary vertex (small DCA) and be isolated; MVTX clusters could not be shared with other tracks. A cluster was searched in ϕ within the same chip (the INTT has coarse z resolution). The efficiency was computed per half-ladder, and a timing study separated hits in the triggered crossing from leakage into the next crossing and from carry-over. Beam tests had shown a very high intrinsic efficiency (public INTT test-beam paper); the data study checks it in the real environment.

Probe impurity (derivation)

Let a fraction P of probe tracks be real, with true efficiency ε, and $1 - P$ be fakes that find a random cluster in the window with probability ε_f (from the hit density times the window size). Then $\varepsilon_{\mathrm{meas}} = P\varepsilon + (1 - P)\varepsilon_f$, so $\varepsilon = [\varepsilon_{\mathrm{meas}} - (1 - P)\varepsilon_f]/P$. Toy: $P = 0.95$, $\varepsilon_f = 0.05$, $\varepsilon_{\mathrm{meas}} = 0.95$ gives $\varepsilon = (0.95 - 0.0025)/0.95 \approx 0.997$. Low measured efficiencies in some half-ladders can therefore be a probe-purity effect, which is why [S28] planned TPC matching to purify the probes. The statistical error is binomial, $\sqrt{\varepsilon(1 - \varepsilon)/N}$, which vanishes as $\varepsilon \to 1$; use Clopper–Pearson intervals there.

Question	Is every INTT half-ladder as efficient in collisions as in the beam test?
Observable	Fraction of probe tracks with a matched inner-layer cluster, per half-ladder.
Data	Streaming p+p, field off.
Method	Tag-and-probe with silicon seeds; fiducial and isolation cuts; residual window; timing analysis.
Key challenge	Probe purity: a fake probe track lowers the measured efficiency.
Lesson	Tag-and-probe is how you can measure single-photon efficiency in data, for example with the $\eta \to \gamma\gamma$ partner search in a tagged sample.

6.3 HCal tower-slope calibration [S42]

Physics and method. Each HCal layer has 1536 towers (24 in η times 64 in ϕ). Towers differ in gain. By azimuthal symmetry, all towers in one η ring should see the same energy spectrum in MB events. The *tower-slope* method fits each tower's spectrum to the summed spectrum of its ring and extracts a relative gain factor. Why this works: if a tower's gain is off by g, its spectrum is the reference stretched by g. For an exponential spectrum $e^{-E/T}$ the measured slope becomes gT; for a power law E^{-m} the counts above a threshold scale as g^{m-1}. The low-energy region is excluded because noise, pedestal shifts, zero suppression and trigger thresholds distort it. [S42] applied this to Run-25 Au+Au outer HCal data in run intervals, looked at the map and distribution of correction factors, and confirmed that the overall calibration from cosmic muons was preserved; the method fixes relative gains, while the absolute scale comes from elsewhere.

Question	Are HCal tower gains uniform and stable in time?
Observable	Tower-to-ring spectrum ratios; correction factor maps per run interval.
Data	Run-25 Au+Au MB; cosmic calibration as reference.
Method	Fit each tower spectrum to its η-ring sum above a noise threshold.
Key challenge	Choosing the fit range; separating gain from local effects (dead neighbours, occupancy differences).
Lesson	This is your Tower Slope method applied to the HCal. Comparing your EMCal MC studies with this data-driven HCal use is a natural capstone component.

6.4 Machine-learning momentum-scale correction [S27]

Physics and method. A wrong magnetic-field map, TPC distortions or misalignment bias the track momentum. Decays with precisely known masses calibrate it: if the reconstructed $K^0_S \to \pi^+\pi^-$ or $\Lambda \to p\pi^-$ mass is shifted, the momenta are wrong. [S27] trained a network that outputs a per-track scale factor $p_T^{\rm corr} = \kappa\, p_T^{\rm reco}$ with $\kappa_q(p_T,\eta,\phi) = 1 + \epsilon(p_T,\eta,\phi) + q\,\delta(p_T,\eta,\phi)$. The charge-independent part ϵ is a scale error; the charge-odd part $q\delta$ is typical of a sagitta bias from misalignment or distortions, which pushes positive and negative tracks in opposite directions. Separating them needs both K^0_S ($\pi^+\pi^-$, symmetric) and Λ, $\bar{\Lambda}$ (asymmetric charge roles). The training maximised an unbinned likelihood of the corrected invariant masses with a peak at the PDG value (an approach adapted from a CMS calibration strategy). Validation: the corrected K^0_S and Λ peaks moved to the PDG mass and narrowed in a run not used in training; $\phi \to K^+K^-$, which uses neither the decay nor kaon tracks in training, also improved; and a closure test with artificial distortions injected into PYTHIA showed that the network recovered them.

Question	Can a learned, charge-dependent momentum scale map remove tracking biases?
Observable	Resonance mass position and width versus p_T, η, ϕ, charge.
Data	p+p track DSTs; PYTHIA with injected distortions.
Method	Neural network regression of κ_q; likelihood of resonance masses; independent-channel validation.
Key challenge	No refit is performed, so the method is limited by track quality; time dependence of corrections.
Lesson	The same idea calibrates the EMCal: π^0 and η peaks constrain a per-tower or per-region energy scale; the η is the independent validation channel.

7 Cross-cutting lessons

7.1 Your own analysis through the template

Apply the six questions to your Run-24 p+p measurement; this is the paragraph a reviewer expects at the top of your analysis note.

Question	How well does perturbative QCD with current fragmentation functions describe π^0 and η production at $\sqrt{s} = 200$ GeV, and what is the η/π^0 ratio? The p+p spectra are also the reference for R_{AA}.
Observable	$E\,d^3\sigma/dp^3$ for π^0 and η at mid-rapidity versus p_T; the ratio η/π^0.
Data	Run-24 p+p, `DST_CALOFITTING`, MB bit 10 and Photon-3/4 bits 29/30; MBD vertex; GL1 scalers; PYTHIA8 with EMCal smearing.
Method	Clusters, diphoton pairs, mixed-event plus residual background, efficiency from simulation, luminosity from scalers and σ_{MBD}.
Key challenge	EMCal energy scale and resolution (yield varies as the scale to the power $m - 1 \approx 6$), trigger turn-on in π^0 p_T, and merging at high p_T.
Lesson	Each case below offers a tool for one of these challenges (see the table in Section 7.2).

7.2 Techniques that recur

Technique	Seen in	Use for your π^0/η analysis
Correlation-breaking background	[S25], [S26], [S38], [S39]	Event mixing; check normalisation window and mixing classes
Two independent methods	[S25], [S26], [S05]	π^0 and η peaks as two scale probes; folding vs direct turn-on
Reference spectrum or run	[S42], [S33]	Tower slopes; run QA against an average run
Tag-and-probe	[S28]	Single-photon efficiency with a tagged partner
Control regions (ABCD)	[S30], [S17]	Purity if you add isolation; background asymmetry studies
Data-driven pure samples	[S29], [S31], [S32]	Conversions and resolved pairs to validate ML or shower cuts
Closure with injected effects	[S27], [S40]	Inject a gain pattern; recover it from peak positions
Feasibility before data	[S29], [S39]	Expected yield per p_T bin before choosing binning
Event-level bias control	[S26], [S34]	Crossing and vertex selections that do not bias the physics

A common failure pattern

Several students reported that something that worked in single-particle simulation needed more work in full events or data: a seed count with the wrong normalisation [S25], a classifier validated only on guns [S31], an O+O centrality that depended on a generator artifact [S26]. The pattern is general. Every step should be tested in three environments, in this order: single particles (does the algorithm work?), full simulated events (does it survive the event environment?), and data control samples (does the simulation describe the detector?).

8 Machine learning in sPHENIX analysis

Goal. Understand supervised learning well enough to build, validate and defend a classifier or regressor in a physics analysis.

8.1 The supervised-learning problem

You have examples $\{(\mathbf{x}_i, y_i)\}$: features $\mathbf{x}$ (shower moments, tower energies, track parameters) and labels y (photon or π^0; a true momentum). A model $f_\theta(\mathbf{x})$ with parameters θ is trained by minimising a loss averaged over the training set, $\mathcal{L}(\theta) = \frac{1}{N}\sum_i \ell(f_\theta(\mathbf{x}_i), y_i)$, usually by stochastic gradient descent: $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$ on small batches, with learning rate η. The aim is not a small training loss but good performance on new data (generalisation).

8.2 Losses come from likelihoods

Classification. Model the probability that $\mathbf{x}$ is signal as $q(\mathbf{x})$. For a label $y \in \{0, 1\}$ the likelihood is $q^y(1-q)^{1-y}$. The negative log-likelihood over the sample is the *binary cross-entropy*,

$$\mathcal{L}_{\mathrm{CE}} = -\sum_i \left[y_i \ln q(\mathbf{x}_i) + (1 - y_i) \ln(1 - q(\mathbf{x}_i)) \right]. \tag{7}$$

For K classes with softmax outputs q_k, $\mathcal{L} = -\sum_i \sum_k p_{ik} \ln q_k(\mathbf{x}_i)$ with one-hot truth p_{ik}; for one sample this is $-\ln q_{\mathrm{true}}$, as used in [S31].

What a well-trained classifier learns

At fixed $\mathbf{x}$, the expected loss is $-[\pi(\mathbf{x}) \ln q + (1 - \pi(\mathbf{x})) \ln(1 - q)]$ with $\pi(\mathbf{x}) = P(y = 1 \mid \mathbf{x})$. Setting the derivative in q to zero gives $q^* = \pi(\mathbf{x})$. With equal class sizes, $\pi = p_s(\mathbf{x})/(p_s(\mathbf{x}) + p_b(\mathbf{x}))$, a monotonic function of the likelihood ratio p_s/p_b. By the Neyman–Pearson lemma, cutting on the likelihood ratio is the most powerful test at fixed background efficiency. So a perfectly trained classifier is optimal, and its output is meaningful only through its ordering and its calibration to the training class fractions.

Regression. If $y = f(\mathbf{x})$+Gaussian noise of width σ, the negative log-likelihood is $\sum_i (y_i - f(\mathbf{x}_i))^2 / 2\sigma^2$: mean-squared error. When no per-object truth exists in data, the loss can be a physics likelihood instead. [S27] used the likelihood of corrected resonance masses under a peak at the PDG value, so the network learns momentum scales from data without truth labels.

8.3 Model families

- **Rectangular cuts.** Transparent and easy to validate; not optimal when features are correlated.
- **Boosted decision trees (BDT).** An ensemble of shallow trees, each fitted to the residuals of the previous ones. Strong on tabular, hand-crafted features; robust and fast. sPHENIX photon ID uses a BDT on shower shapes, and a second BDT rejects non-collision background clusters [S17]. The photon-ID cut was made E_T-dependent and the inputs were chosen to be weakly correlated with isolation, which the ABCD purity method needs.
- **Multilayer perceptron (MLP).** Layers $\mathbf{h}_{l+1} = g(W_l \mathbf{h}_l + \mathbf{b}_l)$ with a non-linear activation g (ReLU, tanh). With enough hidden units it approximates any smooth function.
- **Convolutional network (CNN).** Shares weights across positions of an image, so it learns local patterns (a narrow MIP track versus a broad shower) wherever they appear [S32].
- **Set networks (Deep Sets).** A cluster is an unordered set of towers. The form $f(X) = \rho\big(\sum_{x \in X} \phi(x)\big)$ is invariant under permutations and handles a variable number of towers; "masked sum pooling" ignores padding entries [S31]. Graph networks generalise this by letting towers exchange information with neighbours.

8.4 How a boosted decision tree is built

A decision tree splits the training sample recursively. At each node it chooses the feature and threshold that most reduce an impurity measure, for example the Gini index $G = 1 - \sum_k p_k^2$ of the class fractions p_k in the node (zero for a pure node). A shallow tree is a weak classifier. Gradient boosting builds an additive model $F_M(\mathbf{x}) = \sum_{m=1}^{M} \nu\, h_m(\mathbf{x})$: each new tree h_m is fitted to the negative gradient of the loss with respect to F_{m-1} (for the logistic loss, the residuals $y_i - q_{m-1}(\mathbf{x}_i)$), and $\nu < 1$ is a learning rate that limits overfitting. The output is converted to a probability by $q = 1/(1 + e^{-F})$. The main hyperparameters are the number of trees, their depth, ν, and the minimum node size. Because each tree uses thresholds, BDTs are insensitive to monotonic rescaling of inputs; this is one reason they work well on hand-crafted shower moments.

8.5 Class fractions and calibration

A classifier trained on balanced samples learns $q = p_s/(p_s + p_b)$. In data, signal is rare. If the true signal fraction is π_s, the posterior probability is $q' = \pi_s p_s/(\pi_s p_s + (1 - \pi_s)p_b)$, which follows from q via the odds: $\frac{q'}{1-q'} = \frac{\pi_s}{1-\pi_s}\,\frac{q}{1-q}$. Cutting on q or q' selects the same events (monotonic), but q itself is not the probability that a data cluster is a photon. Purity must be measured (for example by ABCD), not read off the score.

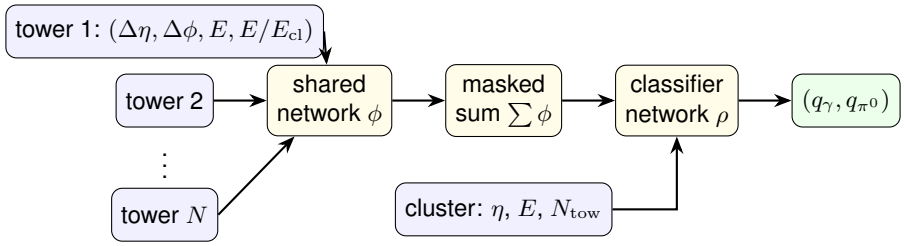


Figure 2 (schematic): a permutation-invariant tower-set classifier of the type described in [S31].

8.6 Training practice

1. **Split** the labelled sample into training, validation (to choose hyperparameters and stop training) and test (used once, for the final performance); 80/10/10 is common [S31].
2. **Balance and reweight.** If signal and background have different p_T or η spectra, the network can learn the spectrum instead of the shower. Reweight one class so that both have the same kinematic distributions, or train in narrow bins.
3. **Scale features** to similar ranges; include only features that simulation models well.
4. **Watch overtraining**: compare score distributions on training and test samples; stop when validation loss stops falling.
5. **Fix the random seeds** and record the software versions so the model can be reproduced for review.

8.7 Measuring performance

For a cut on the score, signal efficiency ε_s and background efficiency ε_b define one point; scanning the cut traces the ROC curve. The area under it (AUC) equals the probability that a random signal scores higher than a random background. The working point is chosen for the analysis, not for the AUC: for a yield measurement, maximise the expected significance $S/\sqrt{S+B}$ or minimise the total uncertainty including the systematic on ε_s.

Worked example (toy)

Before ID, background exceeds signal by a factor 20 ($B/S = 20$). A cut keeps $\varepsilon_s = 0.90$ and $\varepsilon_b = 0.05$. Then $S/B = 0.90/(20 \times 0.05) = 0.9$: purity about 47%. A tighter cut with $\varepsilon_s = 0.70$, $\varepsilon_b = 0.02$ gives $S/B = 1.75$ (purity 64%) but loses 22% of the signal. With $S_0 = 1000$ signal candidates before ID, the significance $S/\sqrt{S+B}$ is $900/\sqrt{1900} = 20.6$ versus $700/\sqrt{1100} = 21.1$: almost equal. The choice is then decided by which efficiency you can validate better in data.

8.8 From simulation to data

A model trained on simulation sees simulated showers. Data differ (noise, calibration, shower-shape mis-modelling; [S17] needed to scale the MC isolation energy to match data). The defence has four parts.

- **Control samples in data** with known labels: photons from conversions $\gamma \to e^+e^-$ and from well-separated $\pi^0 \to \gamma\gamma$ or $\eta \to \gamma\gamma$ pairs; pions from K^0_S; protons from Λ; electrons from conversions [S29, S31].
- **Score comparisons**: the score distribution of a control sample in data and MC must agree; the ratio of cut efficiencies is a scale factor with an uncertainty.
- **Independent validation channels**: test on a process not used in training ($\phi \to K^+K^-$ in [S27]).
- **Closure with injected effects**: distort the simulation in a known way and check that the method recovers it [S27].

Systematic uncertainties of an ML step are then assigned like any other cut (Module 9): vary the working point and apply the Barlow test; use the data–MC efficiency difference in the control sample; retrain with a different generator or seed.

8.9 Tools

TMVA (inside ROOT) trains BDTs and MLPs and applies them in C++ through its reader, which fits a Fun4All module directly. scikit-learn and XGBoost cover classical methods in Python; PyTorch covers networks. Models trained in Python can be exported (for example to ONNX) for C++ inference; check which inference libraries the current `coresoftware` build provides before relying on one. A minimal set-network classifier (toy, PyTorch) looks like this:

```
import torch, torch.nn as nn
class TowerSet(nn.Module):           # per-tower features: deta, dphi, E, E/Eclus
    def __init__(s, nf=4, nglob=3, h=64):
        super().__init__()
        s.phi = nn.Sequential(nn.Linear(nf,h), nn.ReLU(), nn.Linear(h,h), nn.ReLU())
        s.rho = nn.Sequential(nn.Linear(h+nglob,h), nn.ReLU(), nn.Linear(h,2))
    def forward(s, towers, mask, glob):  # towers [B,N,4], mask [B,N], glob [B,3]
        z = (s.phi(towers) * mask.unsqueeze(-1)).sum(1)   # masked sum pooling
        return s.rho(torch.cat([z, glob], 1))             # logits (gamma, pi0)
loss_fn = nn.CrossEntropyLoss()
```

Connect to your analysis

Two ML projects map onto your analysis. (1) A merged-π^0 selector at high p_T, trained on PYTHIA π^0 and single photons with your EMCal smearing, validated on the π^0 peak region at p_T where the photons are still resolved (pick events where the classifier is applied to each photon of a resolved pair). (2) An EMCal energy-scale regression in η and ϕ regions using the π^0 mass as the likelihood target and the η mass as the independent validation, the calorimeter analogue of [S27].

9 Capstone project

Goal. Write a two-page proposal for one improvement to your π^0/η cross-section analysis that uses knowledge from outside the EMCal, and present it to your advisor.

9.1 Template

Use these headings; each is a few sentences.

1. **Title and one-sentence aim.**
2. **Physics motivation**: which uncertainty or limitation of the current analysis it addresses; which factor of the master formula (Module 9, Section 1) it changes.
3. **Current state**: how that factor is obtained now and its estimated uncertainty.
4. **Method**: data, triggers, simulation, reconstruction objects, and the algorithm.
5. **Validation**: closure test in simulation; a data control sample; a cross-check with an independent method.
6. **Expected gain**: a quantitative estimate (toy or simple calculation) of the improvement.
7. **Risks and fallbacks**: what could fail, and the reduced version that still delivers something.
8. **Timeline and deliverables**: milestones over 8–12 weeks; code, an analysis-note section, a TG presentation.

A minimal LaTeX skeleton:

```
\documentclass[11pt]{article}\usepackage[margin=2cm]{geometry}
\begin{document}
\section*{Capstone: <title>}  \textbf{Aim.} ...
\paragraph{Motivation.} ...   \paragraph{Current state.} ...
\paragraph{Method.} ...       \paragraph{Validation.} ...
\paragraph{Expected gain.} ...\paragraph{Risks.} ...
\paragraph{Timeline.} weeks 1-2 ..., 3-6 ..., 7-10 ..., 11-12 write-up
\end{document}
```

9.2 Five worked project ideas

1. **Track-based conversion veto and isolation.** Photons that convert before the EMCal produce $e^{\pm}$ tracks matched to the cluster (track–calo matching, Module 8). Measure the conversion fraction in data versus simulation as a material check; use it to correct the single-photon efficiency. Validation: conversion radius distribution matching known material.

2. **MBD trigger bias for π^0 events.** Measure $\varepsilon_{\mathrm{MBD}}$ for events with a high-p_T π^0 using photon-triggered data in which the MBD requirement was absent (or using ZDC- or clock-triggered samples), and compare with PYTHIA. Changes a global factor of the cross section [S17 shows how large this factor can be for photons].
3. **Streaming-readout luminosity cross-check.** Count MBD coincidences in streaming data per bunch and compare with GL1 scalers; derive the pile-up correction from the bunch-by-bunch rate dependence (Module 9, Section 4.3).
4. **ML merged-π^0 selector** as in the box above, extending the π^0 spectrum to higher p_T; validation on resolved pairs and on the η, which merges at much higher energy.
5. **Run-by-run EMCal scale from tower slopes and π^0 peaks.** Combine the tower-slope method (relative gains, as in [S42]) with the π^0 and η peak positions (absolute scale) in run intervals; quantify the remaining scale uncertainty, which enters the yield with power $m - 1 \approx 6$. Builds directly on your Tower Slope MC studies.

9.3 Self-assessment before you present

- Can you state which number in your error budget changes, and by roughly how much?
- Is there a closure test, and is it non-trivial (independent samples)?
- Is there a data-driven validation that does not use the same assumptions as the method?
- Is the scope achievable in 8–12 weeks with a useful fallback?
- Would a reviewer outside the Cold QCD or Jet TG understand the motivation from the first paragraph?

10 Optional hands-on exercises on SDCC

These are suggestions; the text does not depend on them. Check paths and package availability in the current environment.

1. **Template practice.** Pick a public sPHENIX paper from the references and fill the six-question template in half a page.
2. **Tag-veto toy.** Generate $\pi^0 \to \gamma\gamma$ decays with uniform α; apply your EMCal smearing and a partner threshold; plot the veto efficiency versus π^0 energy and compare with $1 - 2E_{\mathrm{th}}/E$.
3. **BDT versus cuts.** With TMVA, train a BDT on two or three shower moments of single photons and single π^0 from your PYTHIA sample; compare the ROC with rectangular cuts; check overtraining.
4. **Set network.** Train the toy `TowerSet` model on tower lists from the same samples; balance the p_T distributions first.
5. **Scale regression toy.** Inject a known ϕ-dependent 2% gain pattern into simulated towers, reconstruct π^0, and recover the pattern by fitting peak positions in ϕ regions.

11 Self-test questions

Answers are in Appendix A.

1. What do layer rotation [S25, S26], event mixing and like-sign pairs have in common?
2. Why does an over-efficient silicon seeding overcount particles, and what two handles reduce this?

3. The CME observable $\Delta\gamma$ has a background proportional to v_2/N. Explain qualitatively where it comes from.
4. Why is coherent ρ^0 photoproduction concentrated at very low p_T?
5. Why are photon-tagged jets at RHIC quark-enriched?
6. Write the expected J/ψ yield formula and identify which factors also appear in your π^0 cross section.
7. Why do direct photons isolate initial-state spin effects?
8. For $E = 8$ GeV and partner threshold 0.4 GeV, what fraction of π^0 escape a tag veto because the partner is too soft?
9. Show that minimising the expected cross-entropy gives $q^*(\mathbf{x}) = P(y = 1 \mid \mathbf{x})$.
10. Why is a sum over towers used in the set network, rather than concatenating tower features?
11. Why did [S27] need both K^0_S and $\Lambda/\bar{\Lambda}$, and why is $\phi \to K^+K^-$ a good validation?
12. In tag-and-probe, why must the probe be reconstructed without the layer under test?
13. Why does the tower-slope method exclude the low-energy part of the spectrum, and what does it not determine?
14. Name two control samples in data that provide pure photons and one that provides pure pions.

12 Glossary

AUC
Area under the ROC curve; probability that a random signal scores above a random background.

CME
Chiral magnetic effect; charge separation along the magnetic field from a chirality imbalance.

Cross-entropy
Negative log-likelihood loss for classification.

Deep Sets
Permutation-invariant network $\rho(\sum \phi(x))$ over a set of inputs.

Jet charge
$Q_k = \sum q_i (p_{T,i}/p_{T,\mathrm{jet}})^k$.

LDME
Long-distance matrix element in NRQCD factorisation.

QEC
Quarkonium energy correlator.

ROC curve
Signal efficiency versus background efficiency (or rejection) as the score cut varies.

Silicon seed
Track candidate from MVTX and INTT clusters, before TPC matching.

Tag-and-probe
Efficiency measurement using objects selected without the element under test.

Tag veto
Rejection of photon candidates that form a π^0 or η mass with another cluster.

Tower-slope calibration
Relative gain calibration by matching each tower's spectrum to its symmetric neighbours.

Tracklet
Pair of hits in two silicon layers pointing to the vertex.

UPC
Ultra-peripheral collision; photon-induced interaction at large impact parameter.

13 References

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[8] PHENIX Collab., An Upgrade Proposal from the PHENIX Collaboration, arXiv:1501.06197.

[9] sPHENIX Public Results page, `https://www.sphenix.bnl.gov/PublicResults`.

School talks (sPHENIX School 2026, BNL, July 2026): [S04] G. Nukazuka, M. Sarsour, 'Cold QCD TG Analysis Opportunities'; [S05] M. Mitrankova, R. Belmont, 'Bulk TG Analysis Opportunities'; [S17] Y. Go, 'Photon Reconstruction'; [S25] H.-R. Jheng, 'Charged hadron multiplicity in O+O using silicon seeds'; [S26] C.-W. Shih, 'Charged-hadron multiplicity in O+O collisions'; [S27] A. Patton, 'Machine Learning Momentum Scale Correction'; [S28] T. Kikuchi, 'INTT hit detection efficiency study'; [S29] B. Sayki, 'J/psi energy correlators in p+p'; [S30] J. Hwang, 'Transverse single-spin asymmetries of isolated prompt photons'; [S31] R. Koike, 'Neural network discrimination of isolated photons'; [S32] S.-Y. Chen, 'What I have done and what I plan to do next'; [S34] I. Omae, 'Measurement of v_2 in Au+Au'; [S35] R. Takaaze, 'First Step in Jet Analysis'; [S37] J. Singh, 'Constraining the CME using SDM'; [S38] K. Gable, 'Angular correlations of ϕ mesons and high-p_T hadrons'; [S39] L. Mwibanda, 'UPC ρ^0 production in O+O and Au+Au, simulation'; [S40] M. I. Abdulhamid, 'Joint jet mass and charge for quark/gluon jet separation'; [S41] T. Harada, 'Vernier Scan analysis of p+p and O+O in Run 25'; [S42] A. M. Khan, 'HCal Calibration studies'; [S43] J. Park, 'Jet Analysis'.

A Answers

1. Each destroys the physical correlation between the two objects of a pair while keeping the detector acceptance and single-object distributions, so the result estimates combinatorial background.
2. Seeding keeps many candidates per particle (combinatorial and duplicate seeds) to maximise efficiency, expecting TPC matching to remove fakes. Handles: a vertex-association cut on the seed's closest approach to the vertex, and pruning seeds that share clusters.

3. Local charge conservation produces opposite-sign pairs close in azimuth; flow aligns these pairs preferentially with the reaction plane, which gives a positive opposite-minus-same-sign correlation proportional to v_2 and diluted by multiplicity N.
4. In coherent production the photon scatters off the whole nucleus, which stays intact; the momentum transfer is limited by the nuclear form factor to about $\hbar c/R_A$, of order 0.1 GeV.
5. Direct photons are produced mostly by $qg \to q\gamma$, which leaves a quark recoiling against the photon.
6. $N = N_{\mathrm{MB}}(\sigma_{J/\psi}/\sigma_{\mathrm{MBD}})\varepsilon_{\mathrm{MBD}} B_{ee} A\, \varepsilon_{\mathrm{trig}}\varepsilon_{\mathrm{reco}}$. Shared with your analysis: $N_{\mathrm{MB}}/\sigma_{\mathrm{MBD}}$ (luminosity), $\varepsilon_{\mathrm{MBD}}$ for signal events, branching ratio, acceptance, trigger turn-on and reconstruction efficiency.
7. Photons do not fragment and do not interact strongly after production, so there are no final-state (Collins-type) effects.
8. Probability $= 2E_{\mathrm{th}}/E = 0.8/8 = 10\%$.
9. At fixed $\mathbf{x}$ the expected loss is $-[\pi \ln q + (1-\pi)\ln(1-q)]$. Its derivative is $-\pi/q + (1-\pi)/(1-q) = 0$, so $q = \pi$.
10. A sum is permutation invariant and works for any number of towers; concatenation would depend on an arbitrary tower order and a fixed length.
11. K^0_S fixes the charge-symmetric scale ϵ; Λ and $\bar{\Lambda}$ have the heavy and light daughters with opposite charges, which separates the charge-odd term δ. $\phi \to K^+K^-$ was not used in training (neither the decay nor kaons), so it is an independent test.
12. Otherwise the probe would be biased toward events where the tested layer has a hit, and the efficiency would be overestimated.
13. Noise, pedestal shifts, zero suppression and trigger thresholds distort the low-energy region and are not gain effects. The method gives relative gains within a ring, not the absolute energy scale.
14. Photons: conversions $\gamma \to e^+e^-$ (via the pair) and well-separated photon pairs from π^0 or η in the mass window. Pions: $K^0_S \to \pi^+\pi^-$.